

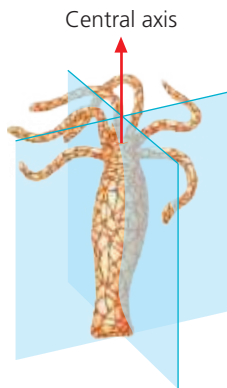
▼ Figure 32.8 VISUALIZING ANIMAL BODY SYMMETRY AND AXES

A basic feature of animal bodies is their type of symmetry. We'll focus on two common types of symmetry: radial symmetry and bilateral symmetry.

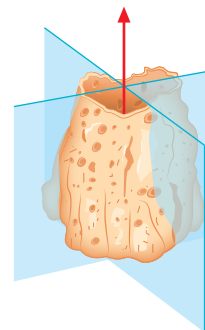
Radial Symmetry

The body parts of a radially symmetrical animal are arranged around a single main axis that passes through the center of the animal. Any imaginary slice through the central axis, as shown here in blue, divides the animal into mirror images.

1. Does a radial animal have a front side and a back side? A left side and a right side? Explain.



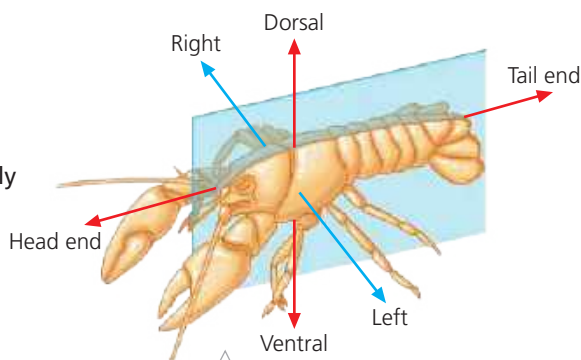
The symmetrical central body cavity of this hydra has a single opening (the "mouth/anus") surrounded by a circular rim of tentacles. Prey captured by the tentacles are pushed into the body cavity and digested.



Many sponges have an irregular shape, lacking symmetry altogether. But like this barrel sponge, some sponges exhibit radial symmetry, with their cells arranged in a regular pattern around a central axis.

Bilateral Symmetry and Body Axes

The body parts of a bilaterally symmetrical animal are arranged around two axes of orientation, the head-tail axis and the dorsal-ventral axis. Only one imaginary slice, again shown in blue, divides the animal into mirror-image halves, referred to as a right side and a left side.



Lobsters and other bilateral animals have a head end and a tail end, a dorsal (top) side and a ventral (bottom) side, as well as a right side and a left side. In nearly all bilateral animals, eyes and other sensory organs are concentrated in the head end of the body.



An upright posture evolved in some groups of bilateral animals, including humans and birds, such as the owl shown here.

2. Draw the two axes of orientation on this owl and label its head end, tail end, and dorsal, ventral, right, and left sides.

➔ **Instructors:** Additional questions related to this Visualizing Figure can be assigned in **Mastering Biology**.

which will help you understand this three-dimensional folding process). As development progresses, these layers, called **germ layers**, form the various tissues and organs of the body. **Ectoderm**, the germ layer covering the surface of the embryo, gives rise to the outer covering of the animal and, in some phyla, to the central nervous system. **Endoderm**, the innermost germ layer, lines the pouch that forms during gastrulation (the archenteron) and gives rise to the lining of the digestive tract (or cavity) and to the lining of organs such as the liver and lungs of vertebrates.

Cnidarians and a few other animal groups that have only these two germ layers are said to be **diploblastic**. All bilaterally symmetrical animals have a third germ layer, called the **mesoderm**, which fills much of the space between the ectoderm and endoderm. Thus, animals with bilateral symmetry are also said to be **triploblastic** (having three germ layers). In triploblasts, the mesoderm forms the muscles and most

other organs between the digestive tract and the outer covering of the animal. Triploblasts include a broad range of animals, from flatworms to arthropods to vertebrates. (Although some diploblasts actually do have a third germ layer, it is not nearly as well developed as the mesoderm of animals considered to be triploblastic.)

Body Cavities

Nearly all animals have a **body cavity**, a fluid- or air-filled space located between the digestive tract (endoderm) and the outer body wall (ectoderm). Body cavities have diverse functions, such as to provide structural support and to facilitate the internal transport of nutrients, gases, and wastes.

Many triploblastic animals have a **coelom** (from the Greek *kóilos*, hollow), a body cavity that forms from tissue derived from mesoderm. The inner and outer layers of mesoderm that surround the cavity connect and form structures that suspend